Received: 24 October 2022 | Revised: 29 November 2022 | Accepted: 11 December 2022

IET Image Processing

DOI: 10.1049/ipr2.12720

The Institution of Engineering and Technology WILEY

ORIGINAL RESEARCH

Real-time video anomaly detection for smart surveillance

Manal Mostafa Ali

Al-Azhar University, Computer and System Engineering, Nasr City, Cairo, Egypt

Correspondence
Manal Mostafa Ali, Al-Azhar University, Computer and System Engineering, Nasr City, Cairo 11751, Egypt.
Email: manal.mustafa@azhar.edu.eg

Abstract

Human monitoring of surveillance cameras for anomaly detection may be a monotonous task as it requires constant attention to judge if the captured activities are anomalous or suspicious. This paper exploits background subtraction (BS), convolutional autoencoder, and object detection for a fully automated surveillance system. BS was performed by modelling each pixel as a mixture of Gaussians (MoG) to concatenate only the higher-order learning in the foreground. Next, the foreground objects are fed to the convolutional autoencoders to filter out abnormal events from normal ones and automatically identify signs of threat and violence in real time. Then, object detection is introduced on the entire scene and the region of interest is highlighted with a bounding box to minimize human intervention in video stream processing. At recognition time, the network generates an alarm for the presence of an anomaly to notify of the identification of potentially suspicious actions. Finally, the complete system is validated upon several benchmark datasets and proved to be robust for complex video anomaly detection. The (AUC) average area under the curve for the frame-level evaluation for all benchmarks is 94.94%. The best improvement ratio of AUC between the proposed system and state-of-the-art methods is 7.7%.

1 | INTRODUCTION

Autonomous video surveillance becomes a fundamental cornerstone in law enforcement, smart monitoring, road safety, environmental monitoring etc. mainly for improving security and public safety [1–4]. It has been considered an inseparable part of crime prevention, smart cities, medical health care, aviation, and natural disasters. One of the most challenging tasks in automated video surveillance is the detection of anomalous events [5]. An anomaly recognition system is described as a software to monitor the indications of offensive or disruptive actions in real-time [6]. Practically, video anomalies are contextual and scene-dependent which means the activity that is anomalous in one scene may be normal in another [7–9]. For instance, riding a bicycle along a bike path is normal behaviour, whereas riding a bicycle down a similar-looking pedestrian sidewalk is considered anomalous.

Considering the massive amounts of videos generated in real-time, manual video analysis by human operators becomes inefficient and expensive [6]. Additionally, monitoring analysts have to wait hours for abnormal occurrences to be captured or seen for immediate reports [3, 10, 11]. The task also becomes monotonous and boring as the occurrence of abnormal events is very minimal when compared to that of normal events [5, 12, 13]. Moreover, events such as traffic accidents, robbery, bullying, shoplifting, violence, and fire in remote places require immediate counteractions to be dealt with properly, which can be facilitated by real-time anomalous detection.

Extraction of meaningful features is one of the key successes of anomaly detection, which can capture the difference between the nominal and anomalous events [1, 4]. Many early anomalous video detection techniques and some recent ones explored the trajectory features such as [14] for vehicle trajectories and [15] for human skeleton trajectory pattern extraction. Other research techniques commonly explore hand-crafted feature extraction such as histogram of optical flow (HOF) [16], and histogram of oriented gradients (HOG) [17]. Recently, due to the successful demonstration of deep learning in computer vision research, several approaches have been proposed for video anomaly detection [1, 18, 19].

This paper presents AVAD (autonomous video anomaly detection) based on deep learning models for the detection of suspicious events in surveillance video. We have filtered uninformative parts of the videos and focused only on the region of the localization object by using BS. Object detection algorithms are employed to obtain the motion of moving objects in the videos. In the event of danger, a detection alarm is triggered, which signals suspicious behaviour on occasion. The method also determines which frame and which parts of it contain the abnormal event to aid the faster judgment of that unusual activity being abnormal or suspicious. This will help the concerned authorities in the prevention of potential catastrophes, meanwhile saving time and labour required in searching the recordings manually.

Accordingly, the contributions are summarized as follows

- This paper proposes a novel framework composed of a semi-supervised deep learning-based feature extraction from video sequences without using any handcrafted features.
- BS is introduced to extract region of interest (ROI) and then the resulting foreground objects are fed into a convolutional autoencoder to distinguish normal and anomalous events.
- Object detector is exploited to provide complete information about anomalous objects along with its confidence score.
- Meanwhile, AVAD can realize real-time warning and recording of an abnormal condition, providing an analytical and amenable decision strategy, then we extensively evaluate the proposed framework on publicly available video anomaly detection data sets.

The remainder of this paper is organized as follows: literature regarding the development of anomaly detection methodologies and state-of-the-art techniques is surveyed in Section 2. Section 3 describes the methodology and key aspects of system architecture. Popular dataset characteristics are summarized in Section 4. Experiments, implementation, and results are proposed in Section 5. Finally, Section 6 presents the conclusion and possible suggestions for future work.

2 | RELATED WORK

Deep learning-based approaches have proved their success in detecting abnormal events from videos, in challenging environments [8, 9, 18, 20, 21]. These techniques can generalize the representations of different objects due to the non-linear transformation performance of learnable models [22]. In the past few years, convolutional neural networks (CNNs) have shown a lot of improvement over state-of-the-art methods due to success of CNNs that can capture the shape and motion-related information in a video sequence [2, 19].

Some researchers have used supervised learning methods which require balanced and prior labelled samples [19]. Obtaining annotations for training is difficult and laborious, specifically for videos [4, 7, 8]. Hence, supervised learning models are not the best-suited choice for video anomaly detection [2, 5, 11]. On the other hand, unsupervised deep learning techniques are more suitable solutions as they aim to learn only normal patterns (the majority of the video) without the need for labelling data [4, 22]. However, this technique results in a high number of false positives as the normal events themselves will be radically changing over time and the latter normal events can be detected as anomalous [10]. To overcome this global issue, weakly or semi-supervised learning approaches were widely investigated as they combine the benefits of supervised methods, without needing exhaustive labelling. Semi-supervised methods for video anomaly detection can be found in [9–11, 24]. The significance of these methods is that they can detect the anomalies using weakly labelled or unlabelled data, where only the normal video samples are given as input for training.

Due to their ability to work with little to no supervision, several works such as [4, 20, 23, 24] specifically demonstrate the effective usage of different types of autoencoders for video anomalous detection. These approaches utilize the raw video frames of normal data as input to the encoder and reconstruct the input frames at the decoder output by learning the normal pattern of frame sequences. However, the computational overhead of reconstructing each raw video frame both spatially and temporally at the decoder output is the major drawback of these methods [11]. Hasan et al. [24] proposed a convolutional autoencoder (Conv-AE) framework for reconstructing the scenes and then computed the reconstruction costs for identifying anomalies. They used the handcrafted motion features that consist of HOG, HOF, and improved trajectory features as input to the autoencoder. Ribeiro et al. [23] reconstruct video frames with the help of Conv-AE and calculate the reconstruction error as an anomaly score. Other fundings proposed by [20] who uses a convolutional spatiotemporal autoencoder to learn the regular patterns in the training videos. Recently ref. [11] combined CNN and bidirectional LSTM (CNN-BiLSTM) autoencoder for anomaly detection in Bank-ATMs environment.

Besides, some researchers turned their attention to other directions such as dictionary learning methods [25], multiple instance learning (MIL) [9], and prediction learning [12, 26]. For instance, Lu et al. [25] proposed a dictionary-based approach to learning normal behaviours and calculated reconstruction errors to detect anomalies. Sultani et al. [9] introduced a broad dataset (UCF) and combined deep learning with MIL to classify real-world anomalies. Prediction learning is attracting more attention, so [26] used a U-Net prediction network and introduced a motion constraint in video prediction by enforcing the optical flow between predicted frames and original frames.

There are various other challenges to video anomaly detection including varying illuminations, complex backgrounds, motion blur, object occlusions, presence of noise etc. that make it one of the challenging tasks in computer vision [11, 27]. Therefore ref. [5] considered variations in illuminations and weather conditions and presented a multi-branch approach equipped with a background learning module and multiple identical branches for detecting anomalies. Furthermore, object detection is essential for further object tracking in video surveillance applications. Some approaches such as ref. [28] utilize object trajectories estimated by object tracking algorithms to understand the nature of the object in the scene and consequently detect outliers.

Motivated by the aforementioned domain challenges, this study aims to design a semi-supervised surveillance system that

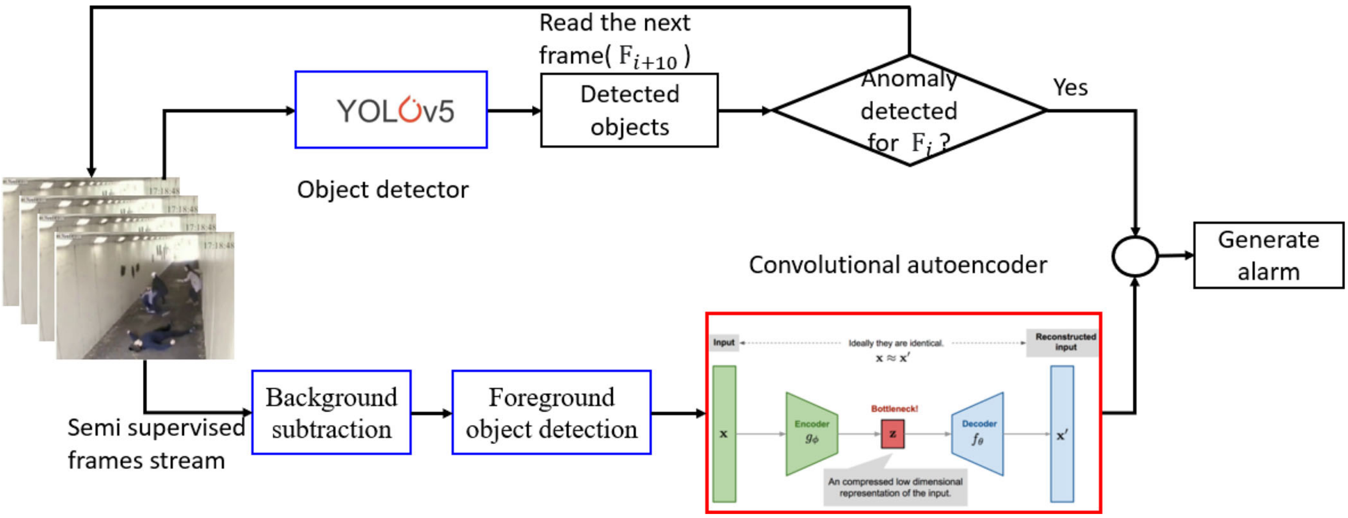

FIGURE 1 Proposed architecture for anomaly detection in surveillance videos

can detect anomalies in the streams of videos as a scalable alternative to manual monitoring. In contrast to full-frame learning, AVAD uses a robust BS for extracting motion, indicating the location of attention regions. The resulting foreground objects are fed into a convolutional autoencoder to distinguish normal and anomalous events. Object detection algorithms were explored to understand the nature of the object in the scene and specifically detect anomalies. Finally, the system generates an alarm if an anomaly appears in the scene.

3 | METHODOLOGY AND SYSTEM ARCHITECTURE

Inspired by [24], we train an end-to-end model that consists of a spatial feature extractor and a temporal autoencoder which together learn the temporal patterns of the frame sequence. As reported by [11] that the reconstruction of the raw frames for training the normal Spatio-temporal pattern at decoder output is computationally expensive. Furthermore, spatial redundant pixels in the frame sequences are less informative and the reconstruction of the raw frames had overhead of reconstructing the redundant information. To overcome these limitations, incorporating a robust BS can help to find the ROI and proceed only with the significant pixels as correctly as possible. BS is used for generating a foreground mask by performing a subtraction between the current frame and a background model, containing the static part of the frame [19].

Dynamic scene changes, repetitive motions (e.g. waiving tree leaves), light reflectance, camera noise, shadows, and sudden illumination variations make reliable and fast object detection difficult. Thus, AVAD uses the fast one-stage method of YOLOv5 as the baseline object detection model. The proposed surveillance system begins by dividing surveillance videos into a fixed number of segments which then undergo four key steps as graphically illustrated in Figure 1 and discussed in the following sections.

3.1 | Background subtraction

A common preprocessing step in computer vision is to discard the background to bring more attention to the foreground objects [5]. over recent years, several methods have been introduced for BS including Mixture of Gaussian (MoG) [29] and the Kernel Density Estimator (KDE) [30]. For pixel-level background model, the background can be obtained by a probability density function (PDF) for each pixel separately. Although a single Gaussian model has been successfully employed for indoor scenes, it is not the best choice for outdoor scenes. For more complex outdoor scenes, several colours can be observed due to complicated scenes and illumination variations [31]. In this case, multinomial PDFs such as MoG [29] have been proposed to model the background by maintaining a PDF per pixel that can cover several values for a single pixel location. Hence, Stauffer's method for MoG is very effective in surveillance systems especially for outdoor scenes because it is adaptive, and can handle multimodal backgrounds. Additionally, it has good performance for stationary and non-stationary backgrounds.

Existing methods for BS can be classified into three main categories: unsupervised [32], supervised [33, 34] and semi-supervised [35]. Unsupervised techniques usually rely on a background model of the video to segment the foreground objects. Supervised methods mainly train very deep CNN to predict the foreground. Finally, semi-supervised approaches can achieve good performance with a small amount of labelled samples. AVAD employs MoG with convolutional autoencoder in a semi-supervised fashion to reduce the effort required to prepare the training set by training the model with a small number of labelled information.

Thresholding is a very popular segmentation technique where each pixel value is compared with the threshold value. Pixel intensities vary between 0 and 255. If the subtracted pixel value is greater than the threshold value, it will take the maximum value (255) for white colour else it will take 0 for the black colour. So, the segmented image gives the moving target in white

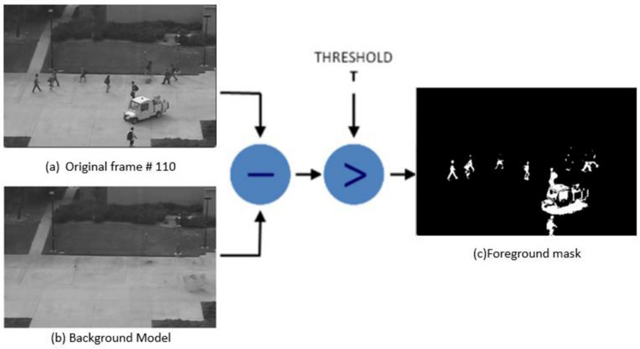

FIGURE 2 Overall structure of BS for UCSD Ped2-Test004 (a) Original frame (b) background model (average frames) and (c) binary image after BS

with a black background, and as result, the moving object can be detected. On OpenCV, the MoG constructor has input parameters that calibrate the behaviour of the method. In practice, several threshold values corresponding to 0.7, 1, and 2 were investigated but the best results were at 0.7. The entire process was indicated in Figure 2 where the frames are presented before and after BS. We can easily see that the moving object has been separated from the background, which greatly concentrates on the ROI for the underlying frames as well as removes unwanted information. Therefore, this approach is very efficient to localize the region before performing anomaly detection through the deep-learning pipeline.

With the depth of information, several challenges including lighting and shadows, noise, and missing parts of objects come to the foreground. As observed from Figure 2, when the colour of the object matches the background then there will exist missing regions that cannot be determined. After several epochs of training, AVAD selects some false-positive samples and maps them into the corresponding locations in the feature map, and subsequentially trains them again. By focusing on the confusing sample training, the model could learn to be more precise and adequately classify again.

3.2 | Foreground object detection

Once the background model is defined, the next step is to identify foreground pixels by comparing the current frame to the background model and then binarizing the difference. The segmentation results obtained by BS are used as a label mask. Then, the foreground objects obtained by BS are used as the input for the convolutional autoencoder. Evidently, it is only needed to judge and classify the foreground object rather than to recognize the entire scene. Therefore, computation can be reduced and processing speed can be significantly improved.

3.3 | Convolutional autoencoders

The main idea of the framework is that the anomalous frames give significantly different motion patterns than normal frames.

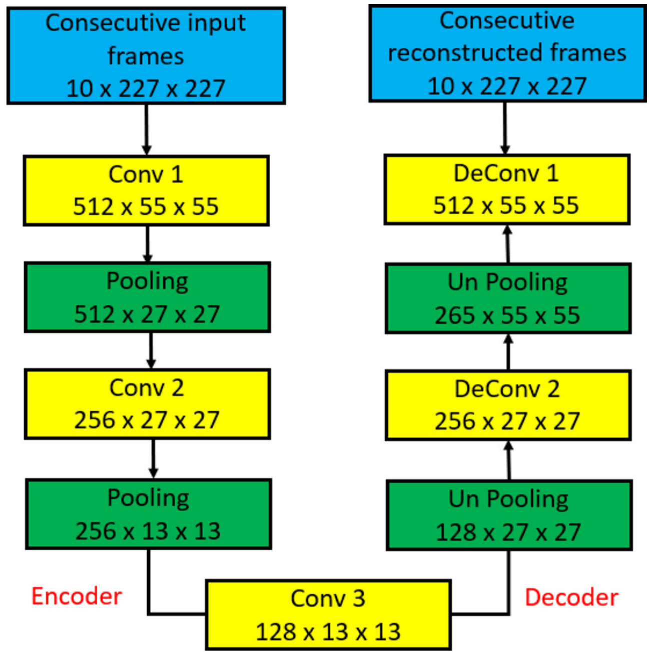

FIGURE 3 Structure of fully convolutional autoencoder

As the model is trained only for the normal pattern of spatiotemporal feature segments, videos with anomalous activities will have a high reconstruction error. For instance, the reconstruction error between the input feature segment and the reconstructed output feature segment is calculated. If the error is beyond a certain threshold, then the underlying video segment can be recognized as abnormal which deviates from the normal ones. Once the autoencoder is properly trained, regular frames are expected to have low reconstruction error, whereas irregular ones consisting of abnormal scenes are expected to have high reconstruction error. As known, the traditional autoencoder networks are incapable of maintaining the spatial features of frames. Therefore, AVAD uses a convolutional autoencoder that learns the optimal filters for minimizing reconstruction error. The architecture of the fully convolutional autoencoder is illustrated in Figure 3.

Spatial convolution maintains the spatial correlation between image areas by using square filters. These filters are used in convolution operations that perform dot products with local regions of a given frame. The most significant issue is restricting an increasing number of filters as it requires extra computational time and memory. The encoder side has three convolutional layers and two pooling layers while the decoder side has three deconvolutional layers and two un-pooling layers. The first convolutional layer has 512 filters with a stride of 4 which produces 512 feature maps with a resolution of 55 × 55 pixels. A kernel size of 2 × 2 pixels had been used for both pooling layers with max poling. The first pooling layer has 512 feature maps of size 27 × 27 pixels. 256 and 128 filters are used for the second and third convolutional respectively. Finally, the encoder produces 128 feature maps of size 13 × 13 pixels. The encoder is for capturing both appearance and motion information of the input video frames while the decoder is for the reconstruction of the

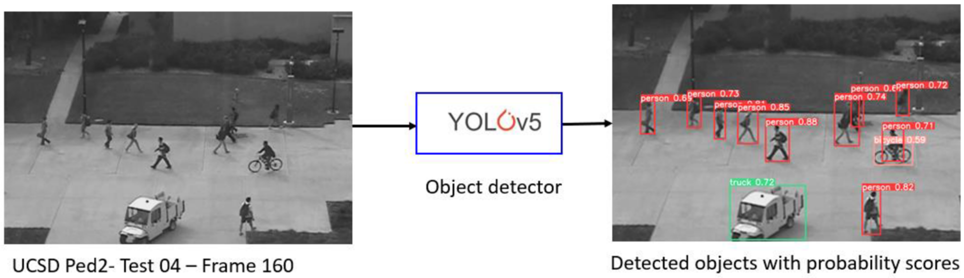

FIGURE 4 Object detection using YOLOv5 on UCSD Ped2-Test 04

frames by deconvolving and un-pooling the input inversely. The output of the final deconvolutional layer is the reconstructed version of the input frames.

3.4 | Object detection

A (YOLO) You Only Look Once [36] was customized to handle the challenges of realistic scenes (e.g. illumination, noise, occlusions, and object scaling) as well as to provide the entire information about the objects in the scene [22]. The pre-trained YOLO is applied on a finite frame interval to predict a set of bounding boxes for the objects. These bounding boxes aim to extract spatial information of the objects from each frame f. Each detected object in every frame has a bounding box (location) along with the class probabilities (appearance). Figure 4 is an example of object detection results on the UCSD Ped2 dataset.

It is well known that consecutive frames in a video stream are likely to be frequently correlated. Since the movement of pedestrians and cyclists is much smaller than cars, there is a high likelihood of a strong temporal correlation between successive frames. Even if the videos contain vehicles, the movement of vehicles on the sidewalk is much less than on the highways. This temporal redundancy can be leveraged to improve the speed of object detection and reduce the computational requirements of object detection. Computationally, applying object detection on every single frame also causes a lot of redundant computation as often two consecutive frames from a video file do not differ greatly. To better utilize consecutive frames, the model was trained with different intervals of input frames corresponding to 5, 10, and 15. We observed that the detection performance shows stability for intervals of both 5 and 10 frames then begin to decrease when applying detection every 15 frames. So, AVAD uses object detection frequency once every 10 frames.

3.5 | Alarm generation

By using this surveillance system, cameras are installed at remote locations and the vision system carefully monitors the live feed generated by these cameras to identify any malicious activity such as fire, stealing, shooting, accident etc. Upon spotting an anomaly, the system automatically alerts the relevant authorities by sending emails, SMS, and raising an alarm. By doing so, AVAD not only aids in preventing crime but also enables the authorities to respond well in time in the face of unwanted events.

4 | CHARACTERISTICS OF VIDEO ANOMALY DATASETS

This section briefly reviews the standard benchmark for evaluating anomaly detection. A summary and characteristics of the most significant datasets are tabulated in Table 1. The first two datasets subway (Entrance and Exit) [37] and unusual crowd activity (UMN) [38] are not recommended for evaluating anomaly detection because of the ambiguities and lack of spatial annotation [7].

UCSD Ped1[1] and Ped2 datasets [13, 39] are the most widely used datasets for video anomaly detection. Both of those datasets had videos from a different static camera observing a pedestrian walkway. UCSD Ped1 includes clips of groups of people walking towards and away from the camera at a low spatial resolution of 158×238 pixels. All non-pedestrian objects and even unusual motions by pedestrians are considered abnormal. The anomalies may be “biker”, “skater”, “cart”, “wheelchair” and “other”. UCSD Ped2 represents scenes with pedestrian movement parallel to the camera plane and contains 16 training and 12 testing video samples of slightly higher resolution, 240×360 pixels.

The CUHK Avenue[2] [25] is mainly about people walking in and out of a building and consists of short videos from a camera looking at the side of a building with pedestrian walkways by it. The anomalous objects may be “throwing papers or bags”, “person runs in front of building”, “wrong direction”, or “child skipping”. Street Scene[3] [40] is the most recent and largest dataset which is taken from a USB camera overlooking a scene of a two-lane street with bike lanes and pedestrian sidewalks during daytime.

All the aforementioned datasets are used for a single-scene video detection formulation. ShanghaiTech[4] [21] and UCF-Crime[5] [9] are useful for multi-scene video anomaly detection. For instance, ShanghaiTech [21] is a recently proposed new

[1] http://www.svcl.ucsd.edu/projects/anomaly/dataset.htm
[2] hhttp://www.cse.cuhk.edu.hk/leojia/projects/detectabnormal/dataset.html
[3] http://www.merl.com/demos/video-anomaly-detection
[4] https://svip-lab.github.io/dataset/campus_dataset.html
[5] http://mha.cs.umn.edu/

TABLE 1 Statistics of the popular benchmark dataset

Benchmark datasets	# of videos	Anomalous events	Setting	Resolution	Average frames	Normal frames	Abnormal frames	Training frames	Testing frames	Total frames
Subway entrance [37]	One long video	66	indoor	512 × 384	121,749	134,124	2400	20,000	116,524	136,524
Subway exit [37]	One long video	19	indoor	512 × 384	64,901	71,681	720	7500	64,901	72,401
UMN [38]	11	11	indoor outdoor	320 × 240	1,290	——	——	1,200	6540	7740
UCSD Ped1 [13]	70	40	outdoor	238 × 158	201	9995	4005	6800	7200	14,000
UCSD Ped2 [13]	28	12	outdoor	360 × 240	163	2924	1636	2550	2010	4560
CHUK Avenue [25]	37	47	indoor outdoor	640 × 360	839	26,832	3820	15,328	15,324	30,652
Street scene [40]	81	205	outdoor	1280 × 720	2509	——	——	56,847	146,410	203,257
ShanghaiTech [21]	437	130	outdoor	856 × 480	726	300,308	17,090	274,515	42,883	317,398
UCF crime [9]	1900	950	indoor outdoor	320 × 240	7247	——	——	11,667,670	2,101,630	13×10^6

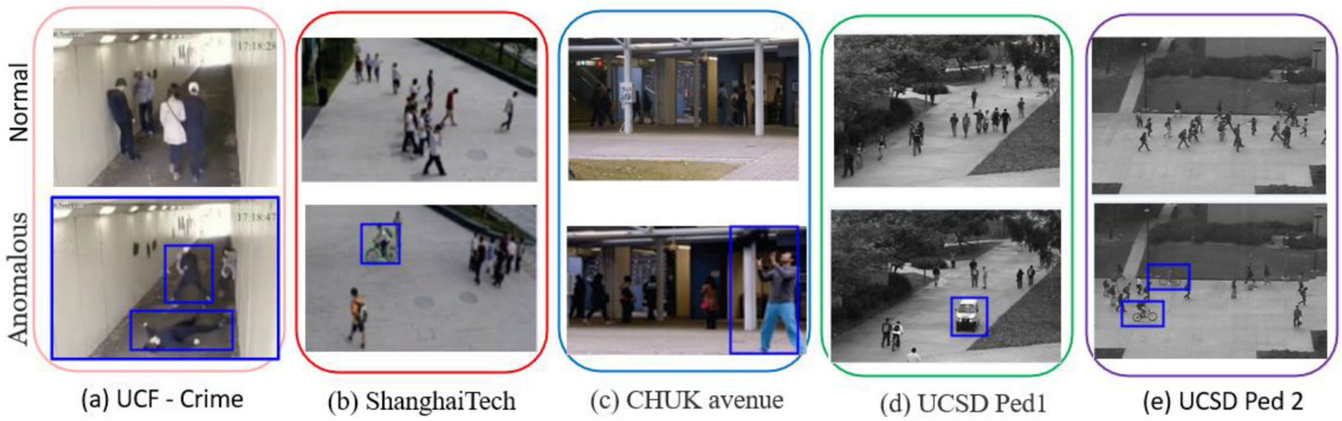

FIGURE 5 Exemplary frames for normal and abnormal scenes from different datasets

dataset for video anomaly detection which contains videos from 13 different scenes. Abnormal activity includes bikers, skateboarders, and people fighting. UCF-Crime [9] contains 128 h of internet videos taken from many different cameras with 13 realistic anomalies including Abuse, Arrest, Arson, Assault, Road accidents etc. Unlike the static backgrounds in ShanghaiTech, UCF-Crime contains complicated and diverse backgrounds. Examples of normal (the top row) and abnormal (the bottom row) frames in the UCF-Crime, ShanghaiTech, CUHK Avenue, and UCSD Ped1&2 datasets are given in Figure 5. The abnormal objects are denoted by blue boxes, such as (a) people who are fighting, (b) a man riding a bicycle on the footpath, (c) a man throwing a bag in front of a building, (d) a vehicle on the wrong direction and (e) people riding bikes on the sidewalk.

5 | EXPERIMENTS AND RESULTS

This section discusses the experimental setup, implementation details, results, and performance analysis of anomaly detection in video surveillance. To identify snippets containing abnormal events, AVAD was validated on five anomaly benchmark data sets, namely UCF-Crime [9], ShanghaiTech [21], CHUK Avenue [25], and UCSD-Ped1&2 [13]. More explanation is presented in the following sections.

5.1 | Experimental setup

All the experiments have been performed on a PC running a 64-bit Windows 10 Pro operating system including an Intel Core i7-7700K CPU and 16 GB DDR4 RAM. The Graphics card was a GeForce GTX 1070 with 256-bit 1920 CUDA cores and 8 GB GDRR5 RAM. Implementations of the models were written in python using OpenCV, Pytorch, Numpy, and Pandas scientific computing libraries. Before being fed into the model, each video is divided into 32 video snippets and each video frame in turn was resized to 240 × 320 pixels for UCF, 192 × 288 pixels for ShanghaiTech, 192 × 320 for CUHK Avenue, 224 × 288 for Ped2, and finally Ped1 remains the same. The intensity of each frame was normalized to [−1, 1] and set the frame rate to 30 fps over all datasets. Given a video sequence as the input, we first estimate the corresponding background scene as anomalies are caused by foreground objects. Mixtures of Gaussians (MoG) are applied to the training frames as an estimation of the background. The predicted background is then subtracted from

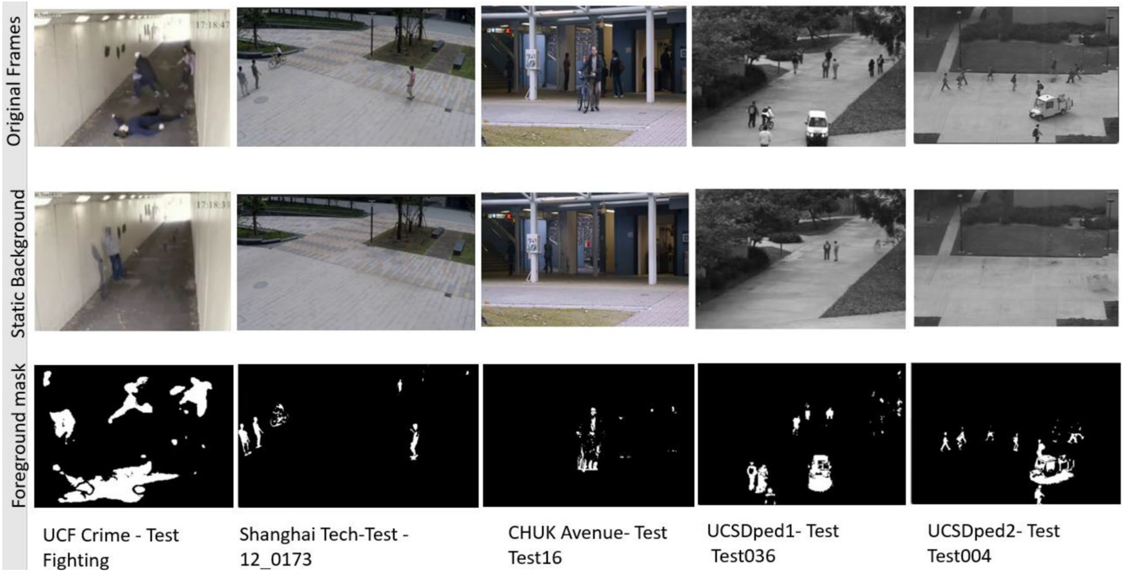

FIGURE 6 Background and foreground prediction for different datasets

TABLE 2 Summary of the AVAD parameters

Parameter	Value
#Videos snippets ∀ video	$T = 32$
#Frames ∀ seconds	30 fs
Frame size	depends on the dataset
Batch size	64
#Epochs	50
An activation function	Sigmoid
Learning rate	10^{-3}, 10^{-4}
A weight decay	5×10^{-4}

the original frame to give the foreground mask. More examples of performing BS are displayed in Figure 6 where every subtraction process for each of the benchmark datasets is indicated by three frames. The first row gives the original frames, the second displays the static background and finally, the last one gives the foreground mask. Hence, only the attention regions of the frame will be fed to the convolutional autoencoder. The weight decay was adjusted at 5×10^{-4} and a batch size of 64 for 50 epochs. The learning rate was set to 10^{-3} for ShanghaiTech and UCF-Crime, and 10^{-4} for CHUK Avenue and UCSD ped 1&2, respectively. Adam optimizer was employed to optimize the reconstruction loss as it allows for setting the learning rate automatically based on the model's weight update history.

To ensure the symmetry of the encoding and decoding, AVAD did not explore the rectified linear unit (ReLU) because activated values from ReLU have no upper bound. Instead, Sigmoid or Hyperbolic tangent had been investigated as the activation function of spatial encoder and decoder. The underlying parameters are summarized in Table 2.

Parameter estimation plays a vital role in machine learning, where we always need to estimate the parameters of the probability distribution. Assume we have independent and identically distributed (IID) random samples $x_1, x_2, \ldots, x_n$ that follow a distribution $f(x_1, x_2, \ldots, x_n; \theta)$, which depends on the unknown parameter θ. The goal is to estimate θ such that it maximizes the probability of observing the underlying random sample. To do this path, we can use Maximum Likelihood (ML) Estimation, Maximum a Posteriori (MAP) Estimation, Minimum Mean Square Error (MMSE) Estimation or Least Square (LS) Estimation. However, for convenience and to focus attention, a heuristic approach is employed rather than parameter estimation.

AVAD uses YOLOv5 for object detection which is based on the Darknet architecture and is pretrained on the MS-COCO dataset [22]. YOLOv5 is presented here because of its high robustness on images in different environments and its optimal speed-accuracy tradeoff. During training, we extract the bounds which have a confidence level greater than 0.5, and for testing, we consider confidence levels greater than or equal to 0.3. Examples of object detectors for UCF—crime, UCSD Ped 1&2 are presented in Figure 7, while examples of CHUK Avenue and ShanghaiTech are given in Figure 8 respectively.

The proposed anomaly detection would be applied to detect a wide range of potential obstacles without providing any additional information about them, such as their class and confidence score. One major difference between anomaly detectors and object detector is that anomaly detectors can identify any objects deviating from the normal pattern but do not provide class labels for the detected objects. Therefore, object detectors are used to provide class labels for those anomalous objects. Together, the anomalous events detected by autoencoder architectures are integrated with the class labels obtained by the object detection model based on YOLOv5 to provide complete information about the anomalous objects.

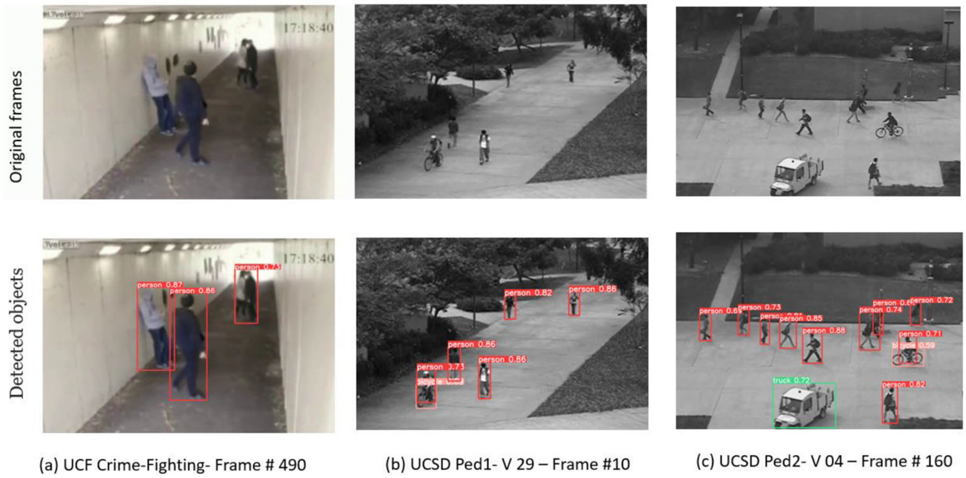

FIGURE 7 Object detection with probability scores for UCF—crime, UCSD Ped 1&2 datasets

FIGURE 8 Object detection with probability scores for CHUK Avenue and ShanghaiTech datasets

Examples of better abnormality detection results along with object detector information are shown in Figure 9. The results show integration of abnormal objects detected by the convolutional autoencoder (in blue colour) and its identical information detected by YOLOv5 (other colours). It is obvious that the anomalies such as burglary (Figure 9a), bikes (Figure 9b), two bikes occurring simultaneously (Figure 9c), motorcycle (Figure 9d) moving on the sidewalk, and intense movements in the front of a building (Figure 9e) are easy to detect. It may be seen that AVAD can detect and separate the moving objects of walking persons, and vehicles almost perfectly.

Some failure cases (surrounded by a purple circle) that impede the performance of our method are reported in Figure 10. Occluded and poorly illuminated frames are difficult to detect as in Figure 10a. A biker is almost the same as a pedestrian from some special angles as in Figure 10b which causes our system to fail to detect it. In Figure 10c, theanomaly is surrounded by normal pedestrians, so AVAD misses it. In this frame, the biker is similar to the grass background and the edge of the bike is too tiny to activate our system. Furthermore, detecting anomalous events with little movement such as a lost package (Figure 10d) is also challenging for the proposed methods. Finally, AVAD cannot detect the child skipping in front of a building as anomalous as in Figure 10e.

5.2 | Evaluation measure

Similar to prior research, to perform a quantitative evaluation, we use both frame-level ground truth and pixel-level ground truth. The first evaluation criterion considers a frame as anomalous if at least one of its pixels is tagged as anomalous. The

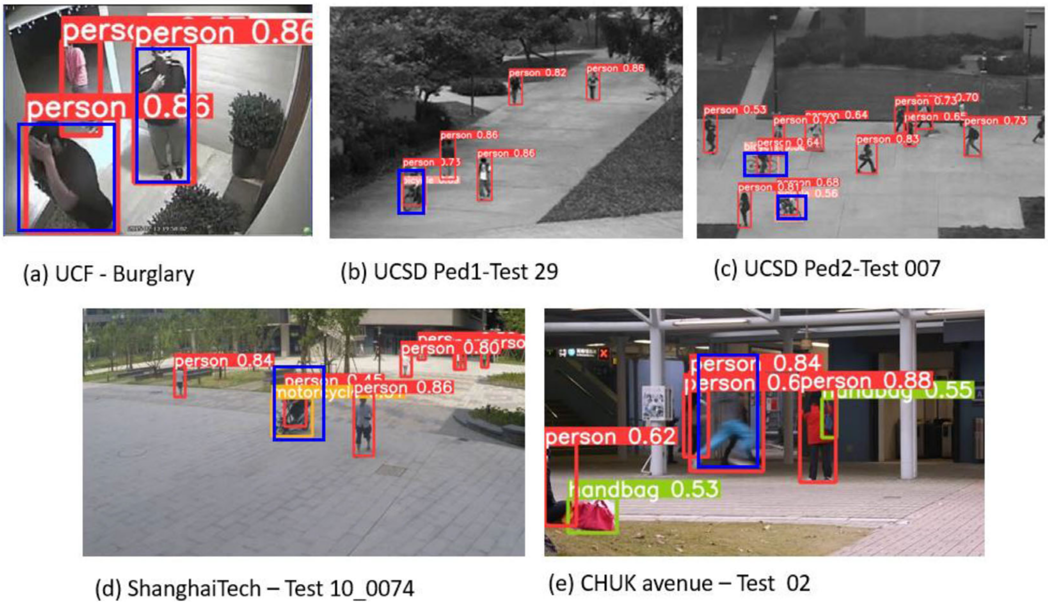

FIGURE 9 Examples of detected anomalous events associated with class label and confidence score

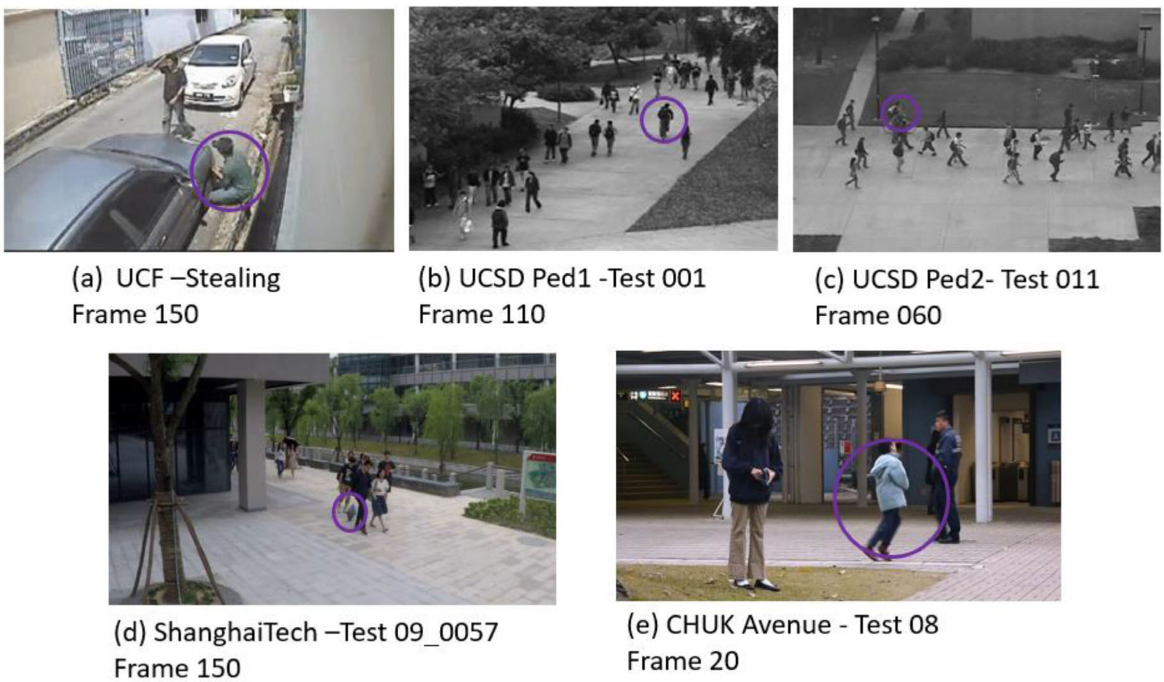

FIGURE 10 Examples of failure abnormality detection results

second criterion considers a frame as anomalous if at least 40% of the anomalous pixels are detected [13]. For both criteria, the area under the curve (AUC) of the receiver operating characteristic curve (ROC) is computed to measure the final performance of the models. The ROC curve is created by plotting the True Positive Rate (TPR) vs the False Positive Rate (FPR) as a function of the detection threshold in the range [0,1], thus summarizing the trade-off between TPR and FPR for a predictive model using different probability thresholds. In practice, AUC of the ROC is used as a measure to determine how the good the model is performing.

5.3 | Results and analysis

All training videos contain only normal events whereas testing videos have both normal and abnormal events. Table 3 provides a performance comparison of the proposed model with

TABLE 3 AUC of AVAD against state-of-the-art methods on UCF, ShanghaiTech, Avenue, and UCSD datasets

			AUC %				
Method	**Algorithm**	**Learning approach**	**UCF crime**	**ShanghaiTech**	**CUHK Avenue**	**UCSD Ped 1**	**UCSD Ped 2**
Hasan et al. 2016 [24]	Conv-AE	Weakly supervised	N/A	N/A	70.2	81.0	90.0
Yong et al. 2017 [20]	ConvLSTM	Weakly supervised	N/A	N/A	80.3	89.9	87.4
Liu et al. 2018 [26]	U-Net	Future frame prediction	N/A	72.8%	85.1%	83.1%	95.4
Wangli et al. 2020 [41]	Temporal Branch ConvNet	Weakly supervised	81.22	**96.74**	N/A	N/A	N/A
Keval et al. 2021 [1]	Generative adversarial net (GAN)	Unsupervised	N/A	70.9	86.4	N/A	**97.2**
Pushpajit et al. 2022 [11]	CNN-BiLSTM-AE	Weakly supervised	91.1	N/A	89.1	N/A	N/A
AVAD	Conv-AE+ BS+ object detection	Weakly supervised	**97.3**	95	**96.8**	**91.8**	93.8

the baseline method proposed in [1, 11, 20, 24, 26, 41], displaying information about their proposed algorithm, the kind of approach (supervised, unsupervised, or weakly supervised), as well as classification quality value the AUC for each proposal. The best performance in each dataset is in bold. The AUC thus represents the degree of separability the model can enforce between normal and anomalous frames. It is worth mentioning that the higher the AUC values, the higher the diagnostic ability of an anomaly detection system.

From the tabulated results, it can be observed that the proposed method generally outperforms the previous methods for the classification of anomalous samples. Specifically, AVAD has achieved higher results on UCF-Crime and Avenue-CHUK datasets respectively. Even though UCF-Crime contains complicated and diverse backgrounds, AVAD achieves an AUC of 97.3%. Whereas the existing state-of-art methods achieve a maximum AUC of 91.1% justifying the outstanding performance of the proposal. For CHUK Avenue dataset, the AUC obtained by AVAD reached 96.8% with an improvement ratio of 7.7% more than the best detection result obtained by [11]. Such improvement gives substantially better sample efficiency and subtle anomaly discriminability than popular prior methods. Further, the results achieved on UCF-Crime and CHUK Avenue datasets show that the AVAD can be applied in both critical indoor environments like banks, and airports as well as real-world outdoor environments like railway platforms, shops etc. for the detection of any anomalous event.

The ShanghaiTech dataset is a large-scale dataset, including over 270K training frames and 42K test frames. Since it has diverse types of normal and anomalous patterns, its performance is relatively lower than those of other datasets. The best performance of AUC was obtained by Wangli et al. [41] at about 96.74% while AVAD gives 95%.

As expected, UCSD Ped1 achieves the lowest performance during experimentation which can be attributed to significant lower resolution (238 × 158) and a lack of consistency in the prior results as some recent works recorded their performance only on a subset of the entire data set [1]. Additionally, some

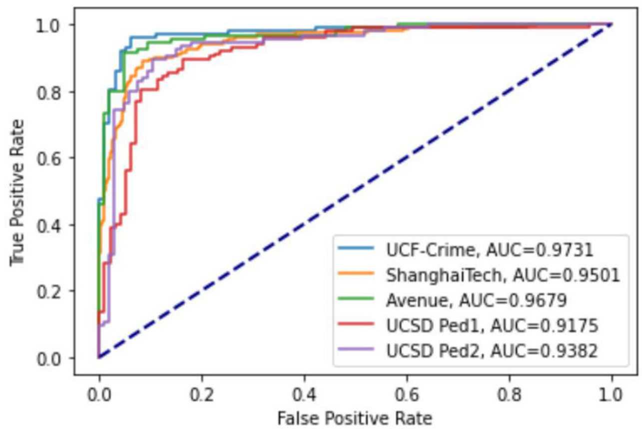

FIGURE 11 Frame-level ROC curves for five benchmark datasets

video sequences have crowd density and caused severe occlusions. However, The AUC on the UCSD Ped1 dataset is 91.8% which increased up to 2% compared to the highest AUC of 89.9% obtained by [20], providing a more accurate location of the abnormal events. Regarding UCSD Ped2, its performance is comparatively reasonable as it achieves a 93.8%. These lower results can be attributed to many reasons including occlusion, poor illumination in some frames, different types of moving objects, and the presence of one or more anomalies in the scene. Moreover, distant abnormal events are indistinguishable and affect pixel-level performance. Additionally, if an anomaly is very close to a normal object, then the abnormal block may be bigger and normal objects are wrongly detected as anomalies. ROC curves for all the benchmark datasets are shown in Figure 11. Such results prove the stability of AVAD over the five benchmark datasets with an average value of AUC reaching 94.94%.

Following these experimental results, it is clear that applying BS, and object detection with the proposed convolutional autoencoder benefits both anomaly detection and localization

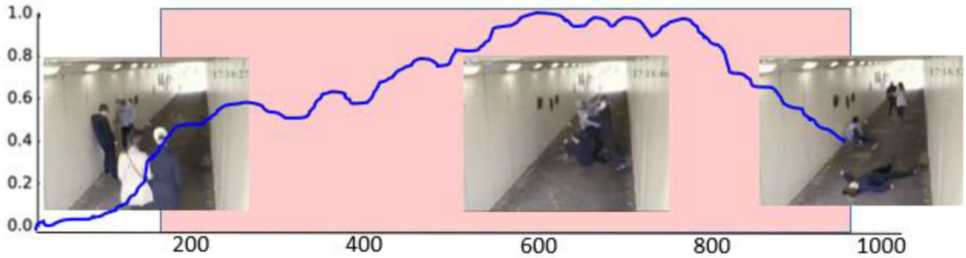

FIGURE 12 Anomaly score of the test video "Fighting" in UCF-Crime dataset

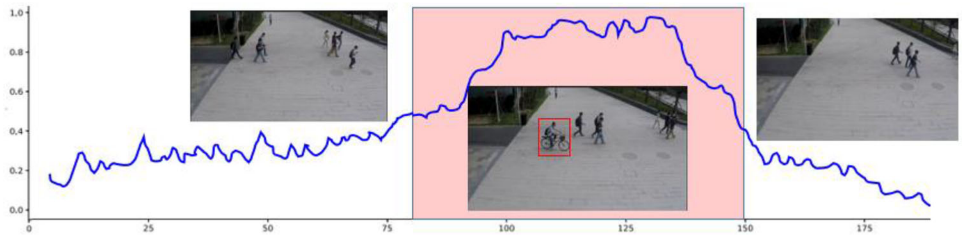

FIGURE 13 Anomaly score of the test video 01 0063 in the ShanghaiTech dataset

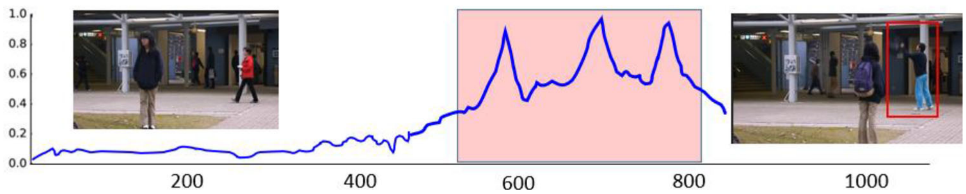

FIGURE 14 Anomaly score of the test video 10 in the CUHK Avenue dataset

tasks. The proposed expert system can realize real-time, robust, and cost-free anomaly detection and localization.

5.4 | Regularity score visualization

AVAD registers the learning regularity for some samples of video sequences as a metric for the qualitative comparison. Figure 12 depicts the learning regularity for test fighting video of UCF crime dataset. It can be observed that when there are irregular motions, the regularity score significantly increases. The regularity score started decreasing gradually when the clash was disengaged and some people took the right direction However, it is still abnormal ($T > 0.5$) because of falling of some persons on the ground. Figure 13 displays regularity score of the test video 01 0063 from the ShanghaiTech dataset where the red rectangle surrounds the abnormal riding object on the footpath. The regularity score of test video 10 from CUHK Avenue dataset is given in Figure 14 where the peek regions indicate appearing of somebody throwing a bag three consecutive times. Figure 15 illustrates the learning regularity score of video 001 in the UCSD Ped1. Increasing regularity scores are associated with appearing of bikes in the wrong direction. Figure 16 draws anomaly score of the test video 02 in the UCSD Ped2 dataset where the red rectangles denote the abnormality of riding a bicycle on the footpath.

The regularity scores are given as a function of frame number where the X-axis refers to the frame number and the Y-axis refers to the regularity score. Overall figures, the learning regularity scores were drawn as a blue line which increased as the irregular motion appears. By doing so AVAD can specifically display which frame and which parts of the recording contain the uncommon activity which helps the quicker judgement of that unordinary action being unusual or suspicious. Besides, it can be seen that AVAD can detect anomalies correctly in these cases even in crowded scenes.

6 | CONCLUSION

Given the important role that video anomaly detection can play in ensuring safety and sometimes prevention of potential catastrophes, one of the main outcomes of video anomaly detection is the real-time decision-making capability. Incorporating a robust BS can help to find the ROI as correct as possible. The activation function chosen to be non-linear so

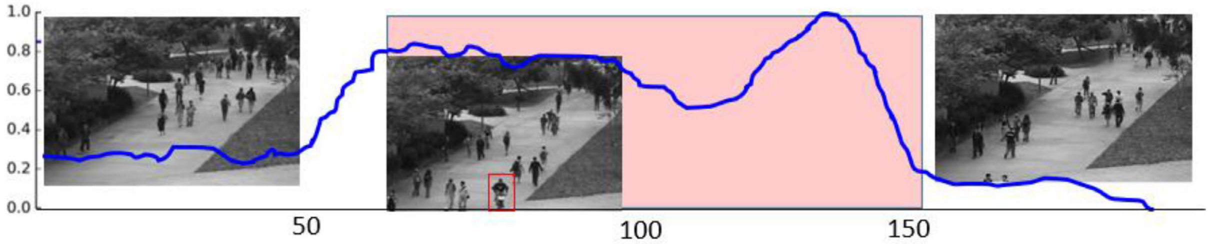

FIGURE 15 Anomaly score of the test video of test video 001 - UCSD Ped1 dataset

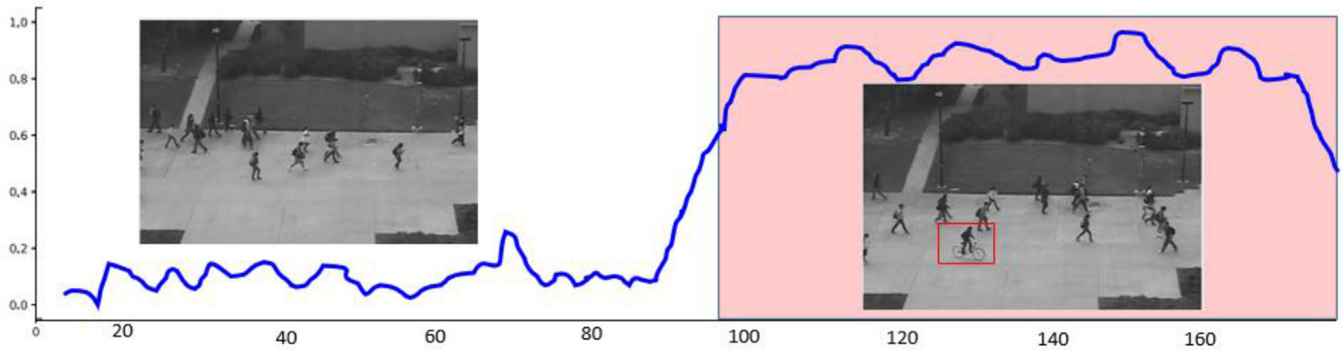

FIGURE 16 Anomaly score of the test video 02 in the UCSD Ped2 dataset

that the autoencoder can extract more useful features than other linear transformation methods. YOLOv5 achieves prominent results during indoor and outdoor environments, under changing illumination conditions and complex backgrounds with clutters. Evaluation measures are calculated to validate the proposed method's performance which includes AUC, ROC, and visualizing regularity scores. Extensive experiments show that AVAD outperforms several state-of-the-art methods by a large margin on most benchmark data sets including UCF-Crime, ShanghaiTech, CHUK Avenue, and UCSD-Ped1&2. In terms of accuracy, AVAD achieves 97.3 accuracies on the best.

From the industrial application point of view, the extraction of the attention region can assist the security officer in focusing on the corresponding anomaly region, instead of a wider, full-framed inspection. There are numerous advantages if such intelligent systems could be realized in our daily lives, such as saving human resources to a large degree and reducing the financial burden on the government. From the practical application point of view, AVAD can be extensively applied in digital education, smart home, public health, and digital twins. Future work should focus more on effective feature extraction methods for improved anomaly event detection using new inputs in an edge computing environment.

AUTHOR CONTRIBUTIONS

M.M. conceived of the main conceptual idea. She developed the theory, performed the computations and verified the analytical methods. She also interpreted the results and worked on the final manuscript.

CONFLICT OF INTEREST

The author certifies that she has no conflict of interest in the subject matter or materials discussed in this manuscript. The author certifies that the article is her original work. The article has not received prior publication and is not under consideration for publication elsewhere.

FUNDING INFORMATION

The author received no financial support for the research, authorship, and/or publication of this article.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available in the public domain at: http://www.svcl.ucsd.edu/projects/anomaly/dataset.htm

http://www.cse.cuhk.edu.hk/leojia/projects/detectabnormal/dataset.html

http://www.merl.com/demos/video-anomaly-detection

https://svip-lab.github.io/dataset/campus_dataset.html

http://mha.cs.umn.edu/

ORCID

Manal Mostafa Ali https://orcid.org/0000-0001-7324-5349

REFERENCES

1. Doshi, K., Yilmaz, Y.: Online anomaly detection in surveillance videos with asymptotic bound on false alarm rate. Pattern Recognit. 114, 107865 (2021)
2. Ristea, N.-C., Madan, N., Ionescu, R.T., Nasrollahi, K., Khan, F.S., Moeslund, T.B., Shah, M.: Self-supervised predictive convolutional

attentive block for anomaly detection. arXiv preprint, arXiv:2111.09099 (2022)
3. Zaheer, M.Z., Mahmood, A., Khan, M.H., Astrid, M., Lee, S.-I.: An anomaly detection system via moving surveillance robots with human collaboration. In: 2021 IEEE/CVF International Conference on Computer Vision Workshops (ICCVW), pp. 2595–2601. Montreal, BC, Canada (2021). https://doi.org/10.1109/ICCVW54120.2021.00293
4. Duman, E., Erdem, A.: Anomaly detection in videos using optical flow and convolutional autoencoder. IEEE Access 7(99), 183914–183923 (2019)
5. Leroux, S., Li, B., Simoens, P.: Multi-branch neural networks for video anomaly detection in adverse lighting and weather conditions. In: *2022 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pp. 3027–3035. Waikoloa, HI, USA (2022). https://doi.org/10.1109/WACV51458.2022.00308
6. Singha, V., Singha, S., Gupta, P.: Real-time anomaly recognition through CCTV using neural networks. In: Procedia Computer Science, International Conference on Smart Sustainable Intelligent Computing and Applications under ICITETM2020. Vol. 173, pp. 254–263 (2020)
7. Ramachandra, B., Jones, M., Vatsavai, R.R.: A survey of single-scene video anomaly detection. IEEE Trans. Pattern Anal. Mach. Intell. 44(5), 2293–2312 (2020)
8. Ren, J.; Xia, F.; Liu, Y.; Lee, I.: Deep video anomaly detection: Opportunities and challenges. In: 2021 International Conference on Data Mining Workshops (ICDMW). Auckland, New Zealand (2021)
9. Sultani, W., Chen, C., Shah, M.: Real-world anomaly detection in surveillance videos. In: IEEE Conference on Computer Vision and Pattern Recognition (CVPR). Salt Lake City, UT, pp. 6479–6488 (2018). https://doi.org/10.1109/CVPR.2018.00678
10. Sarker, M.I., Losada-Gutiérrez, C., Marrón-Romera, M., Fuentes-Jiménez, D., Luengo-Sánchez, S.: Semi-supervised anomaly detection in video-surveillance scenes in the wild. Sensors 21, 3993 (2021). https://doi.org/10.3390/s21123993
11. Khaire, P., Kumar, P.: A semi-supervised deep learning based video anomaly detection framework using RGB-D for surveillance of real-world critical environments. Forensic Sci. Int.: Digital Invest. 40, 301346 (2022)
12. Le, V.T., Kim, Y.G.: Attention-based residual autoencoder for video anomaly detection. Appl. Intell. 52(7), 1–15 (2022). https://doi.org/10.1007/s10489-022-03613-1
13. Li, W., Mahadevan, V., Vasconcelos, N.: Anomaly detection and localization in crowded scenes. IEEE Trans. Pattern Anal. Mach. Intell. 36(1), 18–32 (2014). https://doi.org/10.1109/TPAMI.2013.111
14. Fu, Z., Hu, W., Tan, T.: Similarity based vehicle trajectory clustering and anomaly detection. In: IEEE International Conference on Image Processing 2005, Vol. 2. Genova, Italy (2005)
15. Morais, R., Le, V., Tran, T., Saha, B., Mansour, M., Venkatesh, S.: Learning regularity in skeleton trajectories for anomaly detection in videos. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp. 11996–12004. Long Beach, CA, USA (2019)
16. Colque, R.V.H.M., Caetano, C., de Andrade, M.T.L., Schwartz, W.R.: Histograms of optical flow orientation and magnitude and entropy to detect anomalous events in videos. IEEE Trans. Circuits Syst. Video Technol. 27(3), 673–682 (2016)
17. Dalal, N., Triggs, B.: Histograms of oriented gradients for human detection. In: 2005 IEEE computer society conference on computer vision and pattern recognition (CVPR'05), Vol. 1, pp. 886–893. San Diego, CA, USA (2005)
18. Zhu, S., Chen, C., Sultani, W.: Video anomaly detection for smart surveillance. In: Computer Vision. pp. 1–8. Springer, Cham (2020). https://doi.org/10.1007/978-3-030-03243-2_845-1
19. Vignesh, K., Yadav, G., Sethi, A.: Abnormal event detection on BMTT-PETS 2017 surveillance challenge. In: 2017 IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), pp. 2161–2168. Honolulu, HI, USA (2017). https://doi.org/10.1109/CVPRW.2017.268
20. Chong, Y.S., Tay, Y.H.: Abnormal event detection in videos using spatiotemporal autoencoder. In: Advances in Neural Networks - ISNN 2017. pp. 189–196. Springer International Publishing, Cham (2017)
21. Luo, W., Liu, W., Gao, S.: A revisit of sparse coding based anomaly detection in stacked RNN framework. In: 2017 IEEE International Conference on Computer Vision (ICCV), pp. 341–349. Venice, Italy (2017). https://doi.org/10.1109/ICCV.2017.45
22. Ganokratanaa, T., Aramvith, S., Sebe, N.: Video anomaly detection using deep residual-spatiotemporal translation network. Pattern Recognit. Lett. 155, 143–150 (2022). https://doi.org/10.1016/j.patrec.2021.11.001
23. Ribeiro, M., Lazzaretti, A.E., Lopes, H.S.: A study of deep convolutional auto-encoders for anomaly detection in videos. Pattern Recognit. Lett. 105, 13–22 (2018). https://doi.org/10.1016/j.patrec.2017.07.016
24. Hasan, M., Choi, J., Neumann, J., Roychowdhury, A.K., Davis, L.S.: Learning temporal regularity in video sequences. IEEE Conference on Computer Vision and Pattern Recognition (CVPR). Las Vegas, NV, USA (2016). https://arxiv.org/abs/1604.04574
25. Lu, C., Shi, J., Jia, J.: Abnormal event detection at 150 FPS in MATLAB. In: 2013 IEEE International Conference on Computer Vision, (ICCV), pp. 2720–2727. Sydney, NSW, Australia (2013). https://doi.org/10.1109/ICCV.2013.338
26. Liu, W., Luo, W., Lian, D., Gao, S.: Future frame prediction for anomaly detection–a new baseline. In: IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp. 6536–6545. Salt Lake City, UT, USA (2018)
27. Nasaruddin, N., Muchtar, K., Afdhal, A., et al.: Deep anomaly detection through visual attention in surveillance videos. J. Big Data 7, 87 (2020). https://doi.org/10.1186/s40537-020-00365-y
28. Nikouei, S.Y., Chen, Y., Faughnan, T.R.: Smart surveillance as an edge service for real-time human detection and tracking. In: 2018 IEEE/ACM Symposium on Edge Computing (SEC), pp. 336–337. Seattle, WA, USA (2018). https://doi.org/10.1109/SEC.2018.00036
29. Stauffer, C., Grimson, W.E.L.: Adaptive background mixture models for real-time tracking. In: 1999 IEEE Computer Society Conference on Computer Vision and Pattern Recognition (Cat. No PR00149), Vol. 2, pp. 246–252. Fort Collins, CO, USA (1999). https://doi.org/10.1109/CVPR.1999.784637
30. Elgammal, A., Harwood, D., Davis, L.: Non-parametric model for background subtraction. In: Computer Vision - 6th European Conference on Computer Vision, ECCV 2000, pp. 751–767. Berline, Heidelberg (2000)
31. Al Najjar, M., Ghantous, M., Bayoumi, M.: Object detection. In: Video Surveillance for Sensor Platforms. Lecture Notes in Electrical Engineering. Vol. 114, pp. 97–118. Springer, New York (2014). https://doi.org/10.1007/978-1-4614-1857-3_5
32. Subudhi, B.N., Panda, M.K., Veerakumar, T., Jakhetiya, V., Esakkirajan, S.: Kernel-induced possibilistic fuzzy associate background subtraction for video scene. IEEE Trans. Comput. Social Syst. 1–12 (2022). https://doi.org/10.1109/TCSS.2021.3137306
33. Mandal, M., Vipparthi, S.K.: An empirical review of deep learning frameworks for change detection: Model design, experimental frameworks, challenges and research needs. IEEE Trans. Intell. Transp. Syst. 23(7), 6101–6122 (2022). https://doi.org/10.1109/TITS.2021.3077883
34. Ammar, S., Bouwmans, T., Zaghden, N., Neji, M.: Deep detector classifier (DeepDC) for moving objects segmentation and classification in video surveillance. IET Image Proc. 14(8), 1490–1501 (2020). https://doi.org/10.1049/iet-ipr.2019.0769
35. Giraldo, J.H., Javed, S., Sultana, M., Jung, S.K., Bouwmans, T.: The emerging field of graph signal processing for moving object segmentation. In: Jeong, H., Sumi, K. (eds.), Frontiers of Computer Vision. IW-FCV 2021. Communications in Computer and Information Science. Vol 1405, pp. 31–45. Springer, Cham (2021). https://doi.org/10.1007/978-3-030-81638-4_3
36. Kim, J.-H., Kim, N., Park, Y.W., Won, C.S.: Object detection and classification based on YOLO-V5 with improved maritime dataset. J. Mar. Sci. Eng. 10, 377 (2022). https://doi.org/10.3390/jmse10030377
37. Adam, A., Rivlin, E., Shimshoni, I., Reinitz, D.: Robust real time unusual event detection using multiple fixed-location monitors. IEEE Trans. Pattern Anal. Mach. Intell. 30(3), 555–560 (2008)
38. Unusual Crowd Activity Dataset of University of Minnesota (UMN), Department of Computer Science and Engineering: Monitoring Human

Activity - Home. http://mha.cs.umn.edu/. Retrieved at Aug 17th 2022 (2006)
39. Mahadevan, V., Li, W., Bhalodia, V., Vasconcelos, N.: Anomaly detection in crowded scenes. In: IEEE Conference on Computer Vision and Pattern Recognition (CVPR). pp. 1975–1981. San Francisco, CA, USA (2010)
40. Ramachandra, B., Jones, M.: Street scene: A new dataset and evaluation protocol for video anomaly detection. In: Winter Conference on Applications of Computer Vision (WACV). Snowmass, CO, USA (2020)
41. Hao, W., Zhang, R., Li, S., Li, J., Li, F., Zhao, S., Zhang, W.: Anomaly event detection in security surveillance using two-stream based model. Secur. Commun. Netw. 2020, 8876056 (2020). https://doi.org/10.1155/2020/8876056

How to cite this article: Ali, M.M.: Real-time video anomaly detection for smart surveillance. IET Image Process. 17, 1375–1388 (2023).
https://doi.org/10.1049/ipr2.12720